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SUPERSEDING
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MILITARY SPECIFICATION

STEEL BAR FOR PROJECTILE STOCK

1. SCOPE

1.1 This specification covers cold drawn carbon steel bars no larger than two inches in diameter intended for the manufacture of projectile bodies (see 6.1)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

Federal Test Method Standard No. 151 - Metals, Test Methods

MILITARY

MIL-STD-430A - Macrograph Standards for Steel Bars, Billets and Blooms

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AS.' -A-29 - General Requirements for Hot Rolled and Cold Finished Steel Bars
ASTM-A-370 - Mechanical Testing of Steel Products
ASTM-E-112 - Estimating the Average Grain Size of Metals.

FSC 9510

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(Application for copies of ASTM specifications should be made to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103)

3. REQUIREMENTS

3.1 Material -The finished bars shall be drawn from killed steel produced by the basic open hearth, basic oxygen or electric furnace processes. The ingot from which hot-rolled bars are manufactured shall have come from steel poured into hot-top molds

3.2 Manufacture -Bars shall be cold drawn from hot rolled bars which have been normalized or annealed After cold drawing, all bars shall be stress-relieved to meet the mechanical requirements specified in Table III.

3.3 Chemical composition.-The ladle analysis of each heat shall meet the chemical composition shown in Table I for the grade ordered. (see 6.2)

Table I

Grade	Carbon Percent	Manganese Percent	Phosphorus Percent	Sulfur Percent	Silicon Percent
1030	0.28/0.34	0.60/0.90	0.040 max	0.050 max	0.30 max
1035	0.32/0.38	"	"	"	"
1040	0.37/0.44	"	"	"	"
1045	0.43/0.50	"	"	"	"

The chemical composition of billets or bars, as determined by check analysis, shall meet the chemical requirements shown in Table I within the permissible variations shown in Table II, provided that in each heat of steel the percentage of any element shall not vary both above and below the range shown in Table I

Table II

Permissible Variations in Chemical Composition for Check Analysis

Carbon Percent		Manganese Percent		Phosphorus Percent	Sulphur Percent	Silicon Percent
Over	Under	Over	Under	Over	Over	Over
0.03	0.03	0.03	0.03	0.008	0.008	0.02

3.4 Mechanical properties -Finished bars shall meet the mechanical properties shown in Table III.

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Table III

Mechanical Properties

Bar Size, inch	Yield Strength (Offset 0.2 percent) psi min	Elongation in 2", Min, Percent
1.25 and under	75,000	15
1.26 to 2.0	70,000	15

3.5 Grain size.-The average austenitic grain size shall be No. 5 or finer.

3.6 Soundness.

3.6.1 Macro-etch.

3.6.1 1 Billet -A macroscopic examination of billets shall show no unsound conditions of a nature, degree or extent in excess of the following standards C-3, R-4, S-4 of MIL-STD-430A. This examination shall show no seam, lap, or fold which in the finished bar would exceed the requirement of 3.6.2.

3.6.1.2 Bar.-A macroscopic examination of finished bars shall show no unsound condition of a nature, degree or extent in excess of the following standards: C-3, R-4, S-4 of MIL-STD-430A This examination shall show no seam, lap, or fold to exceed the requirement of 3.6.2. This requirement is not applicable when the steel has met the requirements of 3.6.1.1.

3.6.2 Magnetic.-All finished bars shall be free of defects as detectable by an electromagnetic inspection system approved by the cognizant technical agency. No crack, seam, fold, lap, scabs or slivers shall extend to a depth greater than 0.020 inch below the surface of the bar, except on bars over 1¼ inch in diameter where defects shall be limited to 0.001 inch in depth per 1/16 inch in bar diameter.

3.7 Surface decarburization.-Finished bars shall not be decarburized, partial and total, to a depth greater than 0.020 inch, except on bars over 1¼ inch, which shall be limited to 0 001 inch decarburization per 1/16 inch in bar diameter.

3.8 Dimensions.

3.8.1 Diameter.-The diameter of the finished bar shall be such as to permit the removal from the bar surface during the machining operation, a minimum of 0.026 inch on the radius from bars for 20mm projectiles, 0 040 inch for 30mm projectiles and 0 030 inch for 40mm projectiles. Bars shall be furnished in the size ordered with tolerances on the bar diameter as given in ASTM-A-29, unless otherwise specified. (see 6.2)

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3.8.2 Length.-Finished bars shall be furnished in lengths and to tolerances for length, with the permissible proportion of short lengths, as specified in the procurement document (see 6.2)

3.8.3 Straightness.-Straightness of the finished bars furnished shall be satisfactory for use in automatic screw machines When specific straightness tolerances are required they shall be specified in the procurement document. (see 6.2)

3.8.4 Out-of-roundness.-Out-of-roundness of the finished bars shall be within one-half of the tolerance for the diameter specified in ASTM-A-29.

3 9 Certification -The producer of finished bars shall furnish the purchaser (projectile supplier) a certified report indicating that the steel was manufactured in accordance with this specification This report shall include all data detailed in 4.3.

3.10 Workmanship -Finished bars shall have a smooth surface and shall be uniform in quality and condition within the limits of selective mill practices, and shall be free from hot rolling scale, and other defects which, due to their nature, degree, or extent, will prevent the use of the material for the purpose intended

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

4.2 Lot formation.-A lot shall consist of bars of the same diameter from the same heat of steel and same stress relief furnace charge. In cases where continuous type stress relief furnaces are used, a lot shall consist of all bars of the same diameter from the same heat of steel, stress-relieved in the same run of the continuous type stress relief furnace

4.3 Examination schedule for certification -Records of all examinations and tests on each lot shall be kept complete, and shall be submitted to the Government prior to use of the material. The certification shall include all examinations and tests specified in Table IV

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Table IV

Examinations and Tests	Requirements	Acceptance Criteria	Test Method
Chemical composition	3 3	4 4 1	4.5.1
Mechanical properties	3 4	4.4.2	4.5.2
Grain size	3 5	4.4 3	4.5.3
Soundness			
Macro-etch			
Billet	3 6.1 1	4.4.4.1 1	4 5 4.1
Bar	3.6.1.2	4.4.4.1 2	4.5.4.1
Magnetic	3.6.2	4.4.4.2	4 5.4.2
Surface decarburization	3.7	4.4.5	4 5.5
Dimensions	3.8	4.4.6	4.5.6
Workmanship	3.10	4.4.7	Visual

4.4 Sampling and acceptance criteria.

4.4.1 Chemical composition.

4 4.1 1 Billet -Two check analysis determinations shall be made on each heat of steel. One sample shall be taken from the top end of the top billet of the first usable ingot, and one sample from the bottom end of the bottom billet of the last usable ingot of the heat. Heats of steel which do not meet the chemical composition requirements shall be rejected.

4 4.1.2 Bar -In the event it is necessary to procure bars from steel on which no check analysis has been made, check analysis determinations shall be made on each heat of steel. Samples shall be taken from two randomly selected bars from each 10,000 pounds of steel, or fraction thereof, except that not less than two or more than 10 shall be required for each heat of steel involved. Heats of steel which do not meet the chemical composition requirements shall be rejected

4 4 1.3 Check analysis sampling.-Samples shall be prepared according to method 111.1 of Federal Test Method Standard No. 151A.

4.4.2 Mechanical properties.-Two bars shall be selected randomly from each 10,000 pounds or fraction thereof in each lot, except that not less than 5 nor more than 20 shall represent any lot If any bar fails to meet the requirements of Table III, the lot shall be rejected.

4.4.3 Grain size.

4.4.3.1 Billet.-Two test specimens shall be selected from billets or blooms. One specimen shall be selected from the first usable ingot and one from the last usable ingot of the heat. If the test results show the average grain size to be coarser than the requirement of 3.5, the heat shall be rejected.

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4.4.3.2 Bar.-In the event it is necessary to procure finished bars from steel not inspected in the billet form, a random sample of two bars from each lot shall be examined. If the test results show the average grain size to be coarser than the requirement of 3.5, the lot shall be rejected.

4.4.4 Soundness.

4.4.4.1 Macro-etch.

4.4.4.1.1 Billet.-The billets from each heat intended for application under this specification shall be provided with distinguishing markings as required for the performance of inspection specified herein. The top of the top billet and the bottom of the bottom billet from the first usable ingot, middle ingot and last usable ingot of each heat shall be tested for internal soundness. In addition, the top of the top billet from every intermediate ingot shall be tested, and where necessary, cut back and tested until sound metal is reached Top and bottom refers to the position of the metal in the ingot as poured. No billet will be accepted for use unless a test made from its top end or from a billet above it in the ingot is sound. In the event of failure because of unsoundness of any of the bottom tests, all bottom billets shall be tested and, where necessary, cut back and retested until sound metal is reached.

4.4.4 1 2 Bar.-In the event it is necessary to procure finished bars from steel not inspected in the billet form, bars shall be macro-etch tested to determine internal quality and cleanliness Two bars shall be selected randomly from each 10,000 pounds or fraction thereof in each lot, except that not less than 4 nor more than 20 shall represent any lot. Coupons shall be cut from both ends of the bars selected. If any coupon fails to meet the requirement of 3.6.1.2, the lot shall be rejected.

4.4.4.2 Magnetic -Each finished bar in the lot shall be inspected. Any bar failing to meet the requirement of 3.6 2 shall be rejected.

4.4.5 Surface decarburization -A random sample of two bars from a lot shall be examined for surface decarburization. One coupon shall be cut from one end of each bar. If any coupon shows decarburization greater than the requirement of 3.7, the lot shall be rejected.

4.4.6 Dimensions.-A random sample of 5 bars from a lot shall be selected for dimensional inspection. If any bar fails to meet the requirements for diameter, length, straightness, or out-of-roundness, the lot shall be rejected.

4.4.7 Workmanship.-Each bar shall be visually inspected. Any bar failing to meet the requirement of 3.10 shall be rejected.

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4.4.8 Resubmittal of rejected lots.-A rejected lot may be resubmitted for test if the deficiencies have been corrected or removed. Only one (1) reheat-treatment of each lot is permitted. The resubmitted lot shall comply with all requirements of this specification. However, the retest may be made for only those characteristics which were deficient or which may have been changed during the reworking of the lot.

4.5 Test methods and procedures.

4.5.1 Chemical composition -Tests for chemical composition shall be conducted in accordance with Method 111 1 or 112.1 of Federal Test Method Standard No. 151.

4 5.2 Mechanical properties.-Test for mechanical properties of the finished bars shall be made in accordance with standard tension testing method ASTM-A-370 Replacement of test specimens shall be as specified in Federal Test Method Standard No. 151A.

4.5.3 Grain size.-Grain size will be checked using standard test method ASTM-E-112

4.5 4 Soundness.

4.5.4.1 Macro-etch -The examination of the billet or bar shall be conducted in accordance with Method 321 1, Federal Test Method Standard No. 151.

4.5.4.2 Magnetic -The examination of the bars shall be by test methods and equipment approved by the cognizant technical agency The test system shall incorporate scanning rotation, an automatic indicating or rejection device, and standards for calibration and control, as required by the technical agency.

4.5.5 Surface decarburization.-The depth of the decarburization zone shall be determined by metallographic examination of the entire cross-section of the finished bar. The specimen shall be mounted, polished, etched with 5% nital, and measured at 100 x magnification.

4.5.6 Dimensions.-Sample bars from the lot shall be measured to determine compliance with the requirements of 3.8 using measuring instruments of adequate accuracy.

5. PREPARATION FOR DELIVERY

5.1 Preservation.-Finished bars shall be coated with a corrosion preventative before shipment.

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5.2 Packaging.-All cold drawn bars shall be separated by nominal size and shall be packed and loaded for shipment as specified. Loading and packing shall be such as to insure acceptance by common or other carrier for safe transportation to the point of delivery at the lowest rate.

5.3 Marking -For truck and carload shipments, marking information shall be shown on the bills of lading For bars shipped, boxed, or bundled, each shipping unit shall be marked or tagged with the name of the material, quantity, lot number, specification number, grade number, heat number, diameter, contract or order number and the supplier's name or trademark. For bars shipped loose in less than truck load or carload shipments, each bar shall be stamped or tagged with the lot number and the contract or order number.

6. NOTES

6 1 Intended use.-Finished bars procured under this specification are intended for the manufacture of high explosive or high explosive incendiary projectile bodies by machining or by a combination of cold nosing and machining.

6.2 Ordering data.-Purchasers should exercise any desired options offered herein and procurement documents should specify the following

a. Title, number and date of this specification.

b. Steel grade designation.

c. Diameter of bars.

d. Straightness of bars, if needed

e. Bar length (with permissible proportion of short lengths).

f. Provision for approval of electromagnetic equipment and procedures.

g. Quantity.

h. Packaging - bundle weight.

i. Shipping instructions.

6.3 Ingot pouring.-Ingots known by observation to contain excessive refractory materials should be diverted from the production of bars under this specification.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22 R255

INSTRUCTIONS This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

YES NO *(If "yes" in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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